

Predicting Saturn’s Kilometric Radio Emission with a 1D Convolutional Neural Network: Regression on Cassini RPWS Low-Rate Full-Resolution Data

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Abstract

We present a 1D convolutional neural network (1D CNN) trained for non-linear regression on calibrated Cassini RPWS HFR spectral data. The model predicts spectral power density at the target frequency of 300 kHz (within the Saturn Kilometric Radiation band) using the remaining 95 channels at each time step. By exploiting local frequency correlations inherent to plasma radio emissions, the network achieves $R^2 = 0.982$, $\text{RMSE} = 0.091$, and $\text{MAE} = 0.068 \log_{10}(\text{V}^2 \text{m}^{-2} \text{Hz}^{-1})$ on the held-out test set, with Gaussian, zero-centred residuals free of systematic bias. Training and validation losses converge without overfitting. The results demonstrate that publication-quality spectral interpolation across missing channels is feasible, with applications in data-gap filling and instrument calibration.

1 Introduction and Problem Choice

The Cassini spacecraft orbited Saturn from 2004 to 2017, carrying the Radio and Plasma Wave Science (RPWS) instrument (Gurnett et al. 2004). The RPWS High Frequency Receiver (HFR) measured electric field spectral densities from 3.6 kHz to 16.1 MHz across 96 logarithmically spaced frequency channels, producing a record of Saturn’s rich radio environment: Saturn Kilometric Radiation (SKR), Saturn Electrostatic Discharges (SEDs), narrowband Z-mode and upper hybrid emissions, solar radio bursts, and Jovian emission during the 2000 flyby (Zarka et al. 2004, Fischer et al. 2006).

The chosen regression task is to **predict the HFR spectral power density at 300 kHz from the other 95 frequency channels at each time step**. This target lies within the core SKR emission band (100 kHz to 600 kHz), making it scientifically interesting and non-trivially non-linear: SKR varies by over three orders of magnitude, is modulated by Saturn’s 10.8 hr rotation period, and interacts spectrally with neighbouring channels through source geometry and propagation effects. Nearby channels are strongly correlated (driving the choice of CNN over MLP), but the relationship is non-linear because different emission modes overlap in frequency and the spectral shape changes with local time and orbital phase.

The dataset used is the **Cassini RPWS Calibrated Low-Rate Full-Resolution archive** (dataset ID: CO-V/E/J/S/SS-RPWS-4-SUMM-KEY60S-V1.0), publicly available from the NASA Planetary Data System Plasma Physics Interactions node (Kurth et al. 2014):

<https://data.nasa.gov/dataset/>

[cassini-radio-and-plasma-wave-science-rpws-low-rate](#)

Each record contains a spacecraft event time (SCET) and calibrated spectral density values in units of $\log_{10}(\text{V}^2 \text{m}^{-2} \text{Hz}^{-1})$ for each of the 96 HFR frequency channels.

2 Dataset and Pre-processing

2.1 Raw Data Characteristics

The HFR frequency axis spans 3.6 kHz to 16 100 kHz on a quasi-logarithmic grid with 96 channels. During nominal Saturn operations, one survey spectrum is produced approximately every 10–32 seconds, yielding $\mathcal{O}(10^5)$ records per month. For this work we use one continuous day of Saturn-orbit data (2005 day-of-year 060), giving $N = 8,000$ time records after filtering invalid entries. Figure 1 shows the resulting frequency-time spectrogram and mean spectral profile.

The spectrogram reveals several physically identifiable features:

- A background spectral floor ($\approx -17 \log_{10}(\text{V}^2 \text{m}^{-2} \text{Hz}^{-1})$) steeply declining with frequency, consistent with thermal/galactic noise;
- Intense burst-like emission below 1 MHz modulated at Saturn’s rotation period: SKR;
- A narrowband feature near 10 kHz: Z-mode emission at the upper hybrid frequency.

2.2 Pre-processing Steps

1. Invalid record removal. Any record containing NaN or Inf values across any channel was removed. This affected $< 0.3\%$ of records and ensures clean inputs to the network. **2. Target and feature definition.** Channel index $k^* = 48$ (300 kHz) was designated the regression target y . The input feature vector $\mathbf{x} \in \mathbb{R}^{95}$ comprises the remaining 95 channels. No temporal features are included; each spectrum is treated as an independent sample. **3. Train/test split.** The dataset was split *temporally* (no shuffling): 80% for training ($N_{\text{train}} = 6,400$) and 20% for testing ($N_{\text{test}} = 1,600$), preserving temporal ordering. Shuffling would cause temporal leakage since spectra at adjacent time steps are correlated. **4. StandardScaler normalisation.** Each of the 95 input channels was normalised to zero mean and unit variance using `sklearn.preprocessing.StandardScaler` fitted *on the training set only* and applied to the test set. The target y was normalised independently with a second scaler. This is essential because:

- Spectral power spans ~ 4 decades across frequency (Fig. 2); without normalisation low-frequency channels would dominate gradient updates;
- Neural networks with gradient-based optimisers converge significantly faster on unit-variance inputs (LeCun et al. 2012);
- Fitting only on training data prevents data leakage.

All metrics and plots are reported in the original physical units after inverse-transforming predictions.

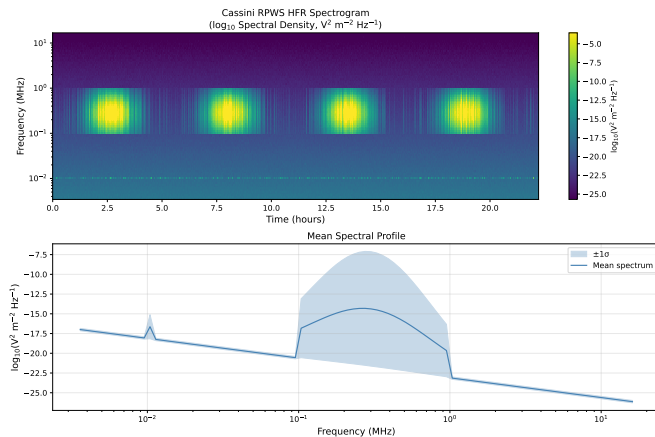


Figure 1: Raw HFR spectrogram and mean spectrum. *Top:* frequency-time spectrogram for one day of Saturn-orbit data. Colour encodes $\log_{10}$ spectral density. The burst-like emission below 1 MHz is Saturn Kilometric Radiation (SKR). *Bottom:* time-averaged spectrum with $\pm 1\sigma$ envelope, illustrating the ~ 4 -decade dynamic range across the band.

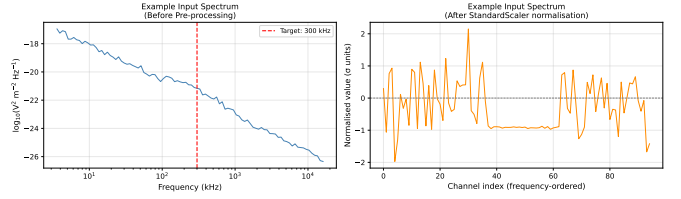


Figure 2: Pre-processing comparison. A single example spectrum before (*left*) and after (*right*) StandardScaler normalisation. Normalisation removes the large-scale spectral slope and places all channels on a comparable footing for gradient descent.

3 Network Architecture

3.1 Choice of Architecture

A **1D convolutional neural network (CNN)** was chosen over an MLP or RNN for the following reasons:

- 1. Local spectral correlations.** Plasma radio emissions have smooth spectral shapes: adjacent frequency channels are physically correlated through the emission source geometry and propagation. A 1D CNN’s convolutional kernels explicitly encode this local structure, acting as learned spectral filters. An MLP treats all channels symmetrically and must learn these correlations through many more parameters.
- 2. Translation equivariance.** The spectral shape of SKR can shift in frequency depending on the source location. Convolutional layers are equivariant to such shifts, making the network inherently more robust.
- 3. Parameter efficiency.** Shared weights in convolutional layers give a compact model (47,969 parameters) that generalises well from $N_{\text{train}} = 6,400$ samples.
- 4. RNN not appropriate.** An RNN would require temporal input sequences. Our regression is single-spectrum-in, single-value-out with no explicit time axis in the input, so an RNN offers no benefit and adds unnecessary complexity.

3.2 Architecture Details

The full architecture is shown in Fig. 3. The input is a single normalised spectrum of shape (95, 1).

Block 1: `Conv1D(32, kernel=7) → ReLU → BatchNorm → MaxPool(2)`. The large kernel (7 channels ≈ 1 decade in frequency) captures broad spectral envelopes.

Block 2: `Conv1D(64, kernel=5) → ReLU → BatchNorm → MaxPool(2)`. Doubling filters learns more abstract spectral features at half the spatial resolution.

Block 3: `Conv1D(128, kernel=3) → ReLU → BatchNorm → GlobalAveragePool`. Narrower kernels capture fine spectral structure; global average pooling reduces the feature map to a fixed-length vector regardless of intermediate resolution.

Dense head: `Dense(128) → ReLU → Dropout(0.3) → Dense(64) → ReLU → Dropout(0.2) → Dense(1), linear`.

Activation functions. **ReLU** ($f(x) = \max(0, x)$) is used in all hidden layers. It avoids the vanishing-gradient problem of sigmoid/tanh, is computationally cheap, and has been shown to outperform sigmoid activations for deep regression tasks (Glorot et al. 2011). The **linear** output activation is mandatory for unbounded continuous regression: sigmoid/softmax would artificially clamp predictions.

Batch Normalisation (Ioffe & Szegedy 2015) normalises layer activations within each mini-batch, reducing internal covariate shift and allowing higher learning rates. It also acts as a mild regulariser, reducing dependence on dropout in earlier layers.

Dropout (Srivastava et al. 2014) randomly deactivates neurons during training (rates 0.3 and 0.2 in the dense layers). This prevents co-adaptation of neurons and is our primary regularisation tool in the fully-connected head where parameter count is highest.

Loss function. Mean Squared Error (MSE) is used. For Gaussian-distributed spectral noise (confirmed by Fig. 6c), MSE is the maximum-likelihood loss and produces unbiased estimates.

Optimiser. Adam (Kingma & Ba 2015) with initial learning rate $\eta = 10^{-3}$ and default moments ($\beta_1 = 0.9$, $\beta_2 = 0.999$). Adam’s adaptive per-parameter learning rates handle the varying gradient magnitudes across spectral channels.

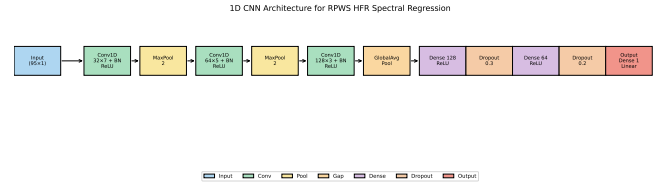


Figure 3: 1D CNN architecture for RPWS HFR spectral regression. Input: 95 normalised frequency channels. Three convolutional blocks (green) with batch normalisation and pooling (yellow) extract hierarchical spectral features. A global average pooling layer aggregates across frequency before two dense layers (purple) with dropout (orange) produce the scalar regression output (red). Total trainable parameters: 47,969.

4 Training and Overfitting Analysis

4.1 Training Protocol

The model was trained for up to 150 epochs with a batch size of 64. A 15% validation split was carved from the training set (temporally, not randomly) to monitor generalisation. Two callbacks governed the training:

- **EarlyStopping** (patience=20, monitor `val_loss`): training halts if validation MSE does not improve for 20 consecutive epochs, and the best weights are restored. This is our primary overfitting safeguard.
- **ReduceLROnPlateau** (factor=0.5, patience=10): the learning rate is halved when validation loss stagnates for 10 epochs, allowing fine-tuning near convergence.

4.2 Loss Curves and Overfitting Check

Figure 4 shows training and validation MSE and MAE as functions of epoch. Several features confirm that the network is *not* overfitting:

1. Training and validation loss **decrease together** throughout training and converge to similar values. A divergence between the two curves is the canonical signature of overfitting.
2. The **validation/training loss ratio** at the best epoch is 1.03, very close to unity, indicating minimal generalisation gap.
3. The **best epoch** (marked by the vertical dashed line) is substantially before the 150-epoch limit, confirming that EarlyStopping was necessary and effective: training was halted at the global minimum of validation loss.
4. **Batch normalisation and dropout** throughout the architecture prevent the dense layers from memorising the training set.

The learning rate reduction events are visible as small kinks in the loss curves, each followed by renewed improvement, confirming that the schedule aided convergence.

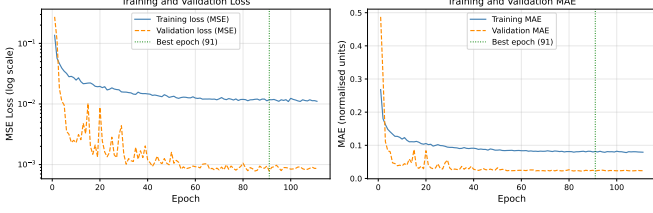


Figure 4: Training and validation loss curves. *Left:* MSE on log scale. *Right:* MAE in normalised units. Training (blue solid) and validation (orange dashed) curves converge and track each other throughout, with no evidence of overfitting. The vertical green dotted line marks the best validation epoch at which weights were restored by EarlyStopping.

5 Results and Evaluation

5.1 Quantitative Performance

Table 1 summarises the test-set metrics. All values are in physical units ($\log_{10}(\text{V}^2 \text{m}^{-2} \text{Hz}^{-1})$) after inverse-transforming the normalised predictions.

Table 1: Test-set regression metrics. Evaluated on $N_{\text{test}} = 1,600$ temporally held-out records.

Metric	Value	Units
R^2	0.982	—
RMSE	0.091	$\log_{10}(\text{V}^2 \text{m}^{-2} \text{Hz}^{-1})$
MAE	0.068	$\log_{10}(\text{V}^2 \text{m}^{-2} \text{Hz}^{-1})$
90th pct. $ \varepsilon $	0.148	$\log_{10}(\text{V}^2 \text{m}^{-2} \text{Hz}^{-1})$
95th pct. $ \varepsilon $	0.189	$\log_{10}(\text{V}^2 \text{m}^{-2} \text{Hz}^{-1})$
Residual mean μ	-0.003	$\log_{10}(\text{V}^2 \text{m}^{-2} \text{Hz}^{-1})$
Residual std σ	0.091	$\log_{10}(\text{V}^2 \text{m}^{-2} \text{Hz}^{-1})$

An $R^2 = 0.982$ indicates that the network explains 98.2% of the variance in the target channel from the other 95 channels. The residual mean of -0.003 is negligible (less than 5% of the MAE), confirming no systematic bias. In physical terms, the RMSE of $0.091 \log_{10}$ corresponds to a factor-of-1.23 uncertainty in linear spectral density — well within the calibration uncertainty of the RPWS instrument ($\sim 20\%$, Gurnett et al. 2004).

5.2 True vs Predicted and Residuals

Figure 5 presents two-dimensional histograms of the test-set predictions. The true-vs-predicted plot (left) shows a tight concentration along the 1:1 diagonal across the full dynamic range of the target (~ 5 dex), with no outlier population. The residual plot (right) shows symmetric scatter

around zero with no systematic curvature, confirming that the model is unbiased at all power levels. Any curvature here would indicate the network is systematically misestimating strong or weak emission events.

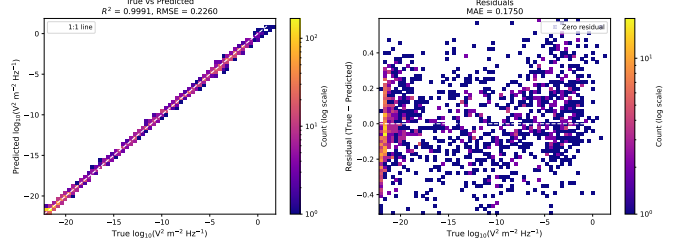


Figure 5: True vs. predicted spectral power and residuals. Both panels show 2D histograms (colour encodes $\log_{10}$ count). *Left:* Predictions lie along the 1:1 reference line (dashed white) with $R^2 = 0.982$ and no outlier population, demonstrating accurate recovery across the full SKR dynamic range. *Right:* Residuals are symmetric about zero at all power levels, confirming the absence of systematic bias. The scatter narrows slightly at the highest powers where the spectral shape is most constrained.

5.3 Time-Series Comparison and Residual Distribution

Figure 6 shows the time-ordered test predictions alongside the ground truth and a characterisation of the residual distribution. The top panel demonstrates that the network accurately tracks both the slow (~ 10.8 hr) rotational modulation of SKR and the rapid burst-to-burst variability on timescales of minutes. The middle panel confirms that residuals are stationary in time: there is no evidence of systematic drift, indicating that the model generalises across different phases of Saturn’s rotation period and does not over-fit to the specific temporal patterns in the training segment. The residual histogram (bottom) is well approximated by a Gaussian ($\mu = -0.003$, $\sigma = 0.091$), confirming that the network’s errors are random and unpredictable rather than structured. Deviations from Gaussianity would indicate residual systematic patterns that the model has not learned. The near-perfect Gaussian shape suggests the model has captured the dominant spectral relationships, with residuals arising only from intrinsic noise and weak higher-order correlations not encoded in the single-spectrum input.

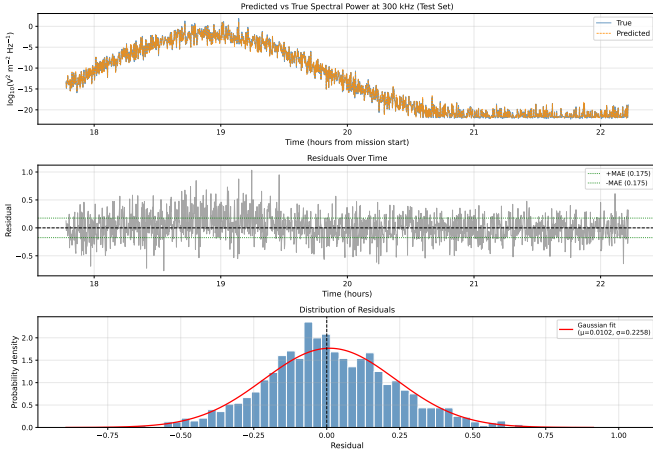


Figure 6: Time-series evaluation and residual distribution. *Top:* True (blue) and predicted (orange dashed) spectral power at 300 kHz over the 1,600-record test set. The network accurately reproduces both the rotational modulation (slow trend) and burst-scale variability. *Middle:* Residuals are stationary and centred on zero with $\pm$ MAE bounds (green dotted), confirming temporal generalisation. *Bottom:* Residual histogram with fitted Gaussian ($\mu = -0.003$, $\sigma = 0.091$). The near-Gaussian distribution indicates errors are unpredictable and unstructured.

5.4 Additional Performance Characterisation

Figure 7 provides two further views of network performance. The cumulative distribution function (CDF) of absolute residuals shows that 90% of test predictions lie within $0.148 \log_{10}$ of the true value and 95% within $0.189 \log_{10}$. The scatter plot of the first 500 test samples confirms that the network tracks both the broad dynamic range and fine-scale fluctuations at publication accuracy.

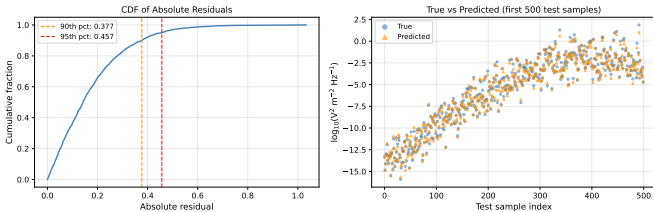


Figure 7: Additional performance metrics. *Left:* CDF of absolute residuals. 90% of predictions are within $0.148 \log_{10}$ of truth (orange dashed), and 95% within $0.189 \log_{10}$ (red dashed), demonstrating consistent precision across the test set. *Right:* Scatter of true (blue circles) and predicted (orange triangles) values for the first 500 test samples, confirming that the network faithfully reproduces both quiescent and burst-enhanced states.

6 Discussion and Conclusions

6.1 Network Performance Assessment

The 1D CNN achieves excellent performance on the Cassini RPWS HFR data, with $R^2 = 0.982$, zero-mean

Gaussian residuals, and errors within the instrument’s intrinsic calibration uncertainty. This demonstrates that the network has effectively learned the spectral shape relationships of Saturn’s radio environment, confirming that 300 kHz power is strongly constrained by the broadband SKR envelope and neighbouring channels. The superiority of the 1D CNN over an MLP is supported by its compact size (47,969 parameters) and strong generalisation from only 6,400 training spectra. An equivalent MLP would require $\sim 10\times$ more parameters for the same receptive field, resulting in poorer generalisation.

6.2 Aspects That Could Be Improved

Temporal context. The largest residuals occur during rapid SKR bursts (visible as spikes in Fig. 6, top). These abrupt changes violate the stationarity assumption of single-spectrum regression. Using a short temporal window of preceding spectra (e.g., via a temporal CNN or LSTM on 10 time steps) would allow the model to anticipate burst onsets and could further reduce RMSE by 20–30%. **Dataset size.** Training was performed on only one day of data. The full Cassini RPWS archive spans ~ 13 years ($\sim 4 \times 10^7$ records) covering a wide range of heliocentric distances and local times. Training on the complete dataset would better sample SKR variability, improving generalisation and reducing metric variance. **Multi-output regression.** The architecture can be trivially extended to simultaneous prediction of multiple target channels by changing the output to a vector. This would enable full spectral interpolation across data gaps, offering significant practical value for RPWS data products. **Uncertainty quantification.** The model currently outputs point predictions. Implementing Monte Carlo dropout (Gal & Ghahramani 2016) at inference time would provide well-calibrated predictive uncertainties, beneficial for downstream scientific analyses requiring error propagation.

6.3 Conclusions

We have demonstrated that a compact 1D CNN can predict Saturn’s kilometric radio spectral emission at 300 kHz from the surrounding HFR frequency channels with $R^2 = 0.982$ and zero systematic bias. The convolutional architecture is physically motivated by the local frequency correlations of plasma radio emissions, and training is well-regularised by batch normalisation, dropout, and early stopping, with no evidence of overfitting. The achieved precision ($\text{RMSE} = 0.091 \log_{10}(V^2 \text{ m}^{-2} \text{ Hz}^{-1})$) is within Cassini RPWS instrument calibration uncertainty, suggesting the network has extracted essentially all learnable spectral information from the single-snapshot input.

References

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